Example 8.22 What is the probability that a leap year selected at random will contain 53 saturdays.

Solution A leap year has 366 days. So it has 52 full weeks and 2 days. 52 Saturdays must be in 52 full weeks.

The possible chances for the remaining two days will be the sample space.

$S = \{\text{Sun-Mon, Mon-Tue, Tue-Wed, Wed-Thu, Thu-Fri, Fri-Sat, Sat-Sun}\}$

$n(S) = 7$

Let A be the event of getting 53rd Saturday.

Then $A = \{\text{Fri-Sat, Sat-Sun}\}$; $n(A) = 2$

Thinking Corner

What will be the probability that a non-leap year will have 53 Saturdays?

Probability of getting 53 Saturdays in a leap year is $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{2}{7}$

Example 8.23 A die is rolled and a coin is tossed simultaneously. Find the probability that the die shows an odd number and the coin shows a head.

Solution Sample space

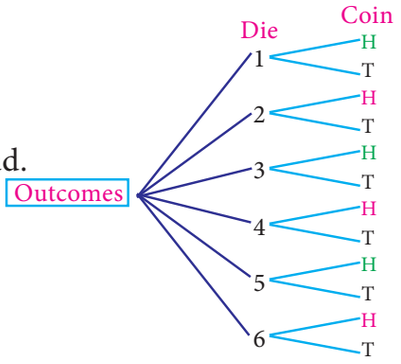
$S = \{1H,1T,2H,2T,3H,3T,4H,4T,5H,5T,6H,6T\}$;

$n(S) = 12$

Let A be the event of getting an odd number and a head.

A $= \{1H, 3H, 5H\}$; $n(A) = 3$

$P(A) = \dfrac{n(A)}{n(S)} = \dfrac{3}{12} = \dfrac{1}{4}$



Activity 3

There are three routes R_1, R_2 and R_3 from Madhu's home to her place of work. There are four parking lots P_1, P_2, P_3, P_4 and three entrances B_1, B_2, B_3 into the office building. There are two elevators E_1 and E_2 to her floor. Using the tree diagram explain how many ways can she reach her office?

Activity 4

Collect the details and find the probabilities of

(i) selecting a boy from your class.
(ii) selecting a girl from your class.
(iii) selecting a student from tenth standard in your school.
(iv) selecting a boy from tenth standard in your school.
(v) selecting a girl from tenth standard in your school.

Example 8.24 A bag contains 6 green balls, some black and red balls. Number of black balls is as twice as the number of red balls. Probability of getting a green ball is thrice the probability of getting a red ball. Find (i) number of black balls (ii) total number of balls.

Solution Number of green balls is $n(G) = 6$

Let number of red balls is $n(R) = x$

Therefore, number of black balls is $n(B) = 2x$

Total number of balls $n(S) = 6 + x + 2x = 6 + 3x$

It is given that, $P(G) = 3 \times P(R)$

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